# TermTrivia: Phase 3 technical report

**CIS 427 Socket Programming Project, final submission (DRAFT)**
Team: Robert and Matthew
Repository: `CIS427_Phase2`, work branch `agentloop/20261029-1650-termtrivia-phase3`
Demonstration: Thursday, November 12th, 12 to 1:45 PM.

**Draft status.** Two sections, the *Team contribution summary* and the *Ripple Report reflection*, are marked **Pending student input** and must be written by the team before submission. All other sections describe the code and the test results recorded in this repository. The team uploads the completed zip to the "P3" folder on Canvas.

## 1. Revised Curiosity Report

# CIS 427 Curiosity Report (Phase 3 revision)

**Project:** TermTrivia, a terminal-based multiplayer trivia game
**Team:** Robert and Matthew
**Source:** `docs/curiosity-report-original.txt` (the Phase 1 report). This revision keeps the original problem, interview evidence, and user stories. It changes only what the built and tested system now shows; each change is marked *Updated* and names the test or measurement behind it.

### A. Problem exploration

**The problem.** Groups who want to play trivia together casually, at a club meeting or a study break, do not have a good lightweight option. Platforms like Kahoot or Quizizz require an internet-connected web client, host account setup, and often a paid tier once you go past a small question limit. The low-tech alternative, one person reading questions out loud, has no real scoring, no persistent leaderboard, and frequent disputes over who answered first.

**Who, when, and how it is handled today.** College students and campus groups in dorm common areas, club meetings, game nights, and study breaks use Kahoot or Quizizz (account, browser, often a paid tier), verbal "shout the answer" games, or index cards. These need internet and an account, or have no scoring that cannot be disputed.

**Why a networked app fits.** Everyone needs to see the same question at the same time and get consistent, fair results: a classic case for a shared server pushing synchronized state to several clients.

### B. Curiosity questions: what testing now tells us

1. **Who would use this most often?** Dorm and friend groups and campus clubs. Still untested with real players: whether they would choose it over Kahoot.
2. **What outcome does the user want?** Fast, fair, fun group trivia with visible scoring. Fairness (correctness-based scoring) is built in and tested: a quick correct answer and a slow

correct answer both score one point (T05, and the demo with a late answer).

3. **What must be shared?** The current question, each player's answer status, and the leaderboard. All three are held in one game object on the server and shown identically to every player in the game.
4. **What must be real time?** Question delivery and results. *Updated:* with 20 healthy localhost clients, the slowest client saw a question within 2 ms of START_ROUND, replies took 1 ms, and the 20-second round timer closed within 6 ms of 20 seconds (`scripts/measure_nfr.sh`).
5. **What if two users answer at nearly the same time?** *Updated:* we tested it. Three clients answering at the same instant were each counted once with correct scores (T36), and two clients joining with the same name at once produced exactly one success and one 409 (T37). Each game has its own lock, so these cases are handled one at a time inside a game without slowing other games.
6. **What would make a user stop trusting the app?** A wrong or laggy leaderboard, a crash when a player disconnects, or not knowing whether an answer registered. Every command gets exactly one reply, a dropped player never crashes the server (T46), and a stalled player in one game does not delay another game (T40).
7. **What is the minimum valuable feature set?** Create or join a game by code, the host starts it, players answer, and the leaderboard updates after each round. All of it works with several clients at once.
8. **What changes from 5 players to 25?** *Updated:* we measured 20 players in one game. Broadcast time did not grow noticeably on localhost. We did not test 25 players, or players on a real network. See the scalability analysis in the technical report.
9. **Does anything persist after a disconnect?** Yes. A disconnected player can reconnect within 60 seconds and keep their score (T46). *Updated:* if the *host* does not return, a started game continues to its normal end and a lobby ends for everyone (T42, T47).
10. **What accessibility barriers exist?** A command-line interface favors people comfortable with a terminal and typing. Speed is never part of scoring, which reduces the penalty for slow typists, but a question still closes after 20 seconds.

### C. User and stakeholder evidence

**Source.** Short interviews conducted with 2 classmates who currently play trivia in group settings.

**What we learned.** Both interviewees said that when they play group trivia, they typically use Kahoot. The main downside they raised was that several useful features, including creating your own games, require either payment or an account, which adds friction for a casual, one-off trivia session with friends. When we described the correctness-only scoring approach (full credit for a correct answer regardless of how fast it was submitted), both said they preferred it over speed-based scoring, since it rewards actually understanding the question rather than reaction time.

**How this shaped the proposal.** This evidence reinforced two core decisions: no-account, no-cost access and correctness-only scoring. Both are unchanged in the finished system. We have not run new interviews in Phase 3, so no new user evidence is claimed.

### D. Target users

**Primary users:** hosts and players in casual trivia settings: dorm groups, campus clubs, friend hangouts. The host is also a participant and answers alongside everyone else. **Secondary stakeholders:** club organizers or RAs running a quick group activity. **Goals:** a fair, low-friction session with no account and no cost. **Likely frustrations:** input lag, uncertainty about whether an

answer registered, disputes over who answered first, long gaps between rounds. **Assumptions about experience:** comfort with a terminal is likely for a CS-adjacent test group and less certain for a general dorm crowd; this remains a real limitation.

**User stories.** As a group organizer, I want to start a game and share a join code so that everyone can jump in quickly with no installs or accounts. As a player, I want to see my score update after each question so that I know how I am doing. As a host, I want to see each round's results and the correct answer after it closes so that I can keep the game moving.

### E. Value proposition

For casual campus groups who need a fast, fair way to play trivia together, TermTrivia provides a lightweight, terminal-based multiplayer quiz game joinable with a simple code. Unlike Kahoot or Quizizz, which require a web client, account setup, and often a paid tier, TermTrivia runs over a direct connection with no sign-up and no cost.

- **Functional value:** synchronized question delivery, fair scoring, and a live leaderboard, for several games at once.
- **Experiential value:** join in seconds with just a code; no install of anything but the client, no account.
- **Reliability value:** scoring depends only on correctness; a disconnected player can reconnect with their score and does not take down the round for everyone else (T46).
- **Scalability value:** *Updated:* the design target of about 20 simultaneous players was tested with 20 players on one machine and met the timing limits. Larger crowds and real networks remain untested.

### F. Assumptions, risks, and open questions: status

1. **Assumption: terminal comfort.** Unchanged and still unverified beyond the two interviewees.
2. **Assumption: about 20 players.** *Updated:* tested with 20 on localhost.
3. **Assumption: same network.** Unchanged; all testing used localhost.
4. **Risk: a mid-game joiner starts at 0.** *Updated:* the joiner waits for the next question and sees no missed questions (T41). They still start at zero; we did not change that.
5. **Risk: telling a reconnect from a new join.** Resolved: reconnecting needs name, game code, and a private token, and every wrong guess gets the same 410 (T28).
6. **Risk: lag with about 20 clients.** *Updated:* not observed on localhost (replies 1 ms, question push 2 ms); a real network could differ.
7. **Open: how long to wait for a disconnected player?** Decided: 60 seconds by default, configurable.
8. **Open: show a mid-game joiner the missed questions?** Decided: no, they start at the next question. A join in the final intermission is also accepted and shown in the final results with score zero (T43).

## 2. Final problem statement and value proposition

**Problem.** Casual groups (dorm friends, campus clubs) lack a lightweight way to play fair, scored trivia together without accounts, payment, or a browser. Reading questions aloud has no reliable scoring or leaderboard and invites arguments about who answered first.

**Solution.** TermTrivia is a client-server terminal game over TCP. One server hosts many games at once. A host creates a game with a join code, players join with the code, and after START_ROUND the server runs the whole game on timers: a 20-second question, a 5-second intermission, and a 10-second final intermission. Scores come from correct answers only. A player who disconnects can reconnect within 60 seconds with a private token and keep their score.

**Value proposition.** For casual campus groups who need a fast, fair way to play trivia together, TermTrivia provides a lightweight, terminal-based multiplayer quiz game joinable with a simple code. Unlike Kahoot or Quizizz, it needs no account, no web client, and no payment. Its distinctive feature is **correctness-only scoring**, which two interviewed classmates preferred to speed-based scoring and which keeps the game fair for slow typists.

## 3. Final requirements

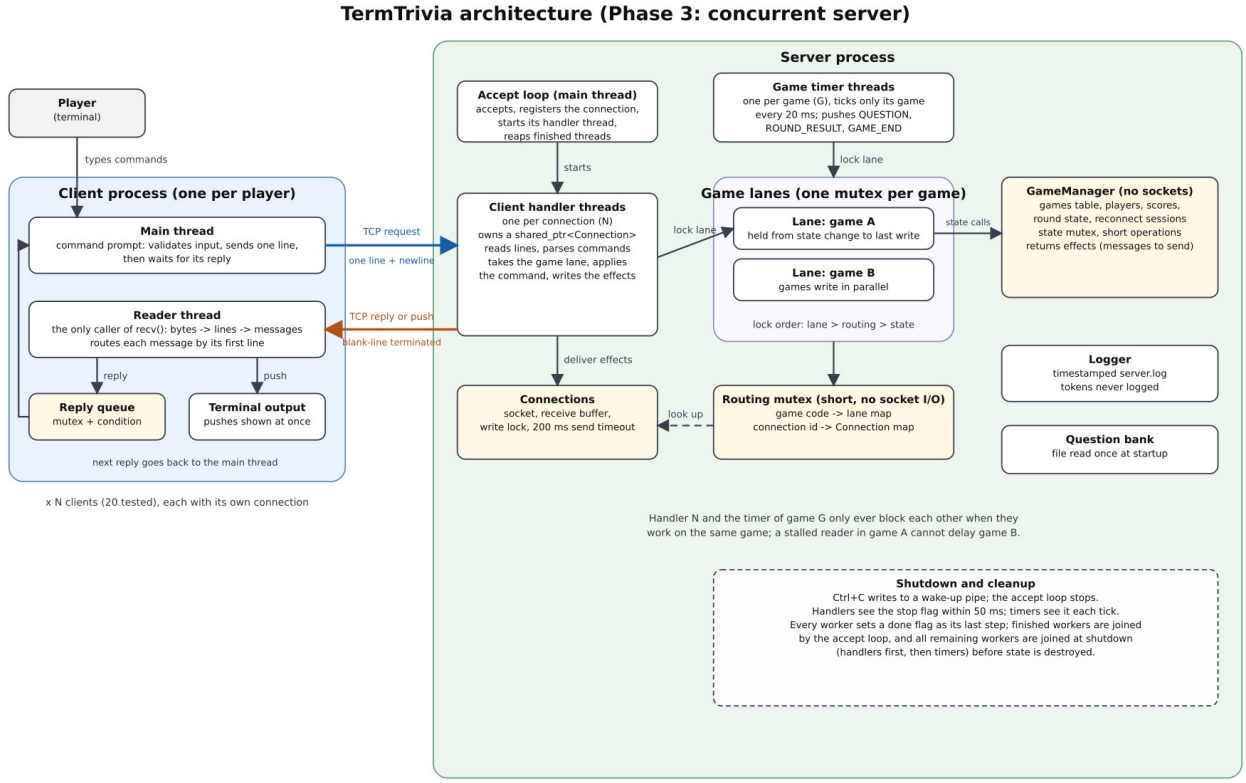
The full text, with protocol references and the tests that verify each, is in `docs/requirements.md` (26 functional and 11 nonfunctional requirements). Summary:

| ID | Requirement (short) |
|---|---|
| FR-01 to FR-03 | Create a game and show its join code; join by code, in the lobby, mid-game, or in the final intermission; the host starts the game once and everything after is automatic. |
| FR-04 to FR-09 | Questions pushed to every player at once; answer A to D; scoring on correctness alone; a round closes when all active players answered or after 20 s; leaderboard on request; automatic end with GAME_END COMPLETE. |
| FR-10, FR-11 | Host END_GAME ends only that game; QUIT is permanent, a non-host quit removes only that player and a host quit ends the game. |
| FR-12, FR-21, FR-22 | Unexpected disconnect keeps the player for 60 s; reconnect with name, code, and token restores the score; the game does not stop when the host drops. |
| FR-13 to FR-20 | Specific error replies (300, 401, 402, 404, 409, 410, 420, 421, 422, 423); connection stays open after a bad line; client handles bad input and server loss. |
| FR-23 | A separate score for every player in every game; several games at once without interference. |
| FR-24, FR-25 | Messages rebuilt from any split or combined reads; timestamped server log. |
| FR-26 (new in Phase 3) | Host expiry: a lobby whose host does not return ends with GAME_END HOST_EXPIRED; a started game continues to COMPLETE with no host. |
| NFR-01 | Up to 20 players: replies within 500 ms, questions within 1 s, timers within 1 s. |
| NFR-02 to | No crash on bad input or disconnects; reconnect rules; correct framing at any split. |

| | |
|---|---|
| NFR-04 | |
| NFR-05 to NFR-06, NFR-10 | Usability, instant display of pushes, scoring independent of speed. |
| NFR-07 | At least 3 simultaneous clients and several simultaneous games. |
| NFR-08, NFR-09, NFR-11 | Separated modules, no tokens in output or logs, 512-byte lines, configurable ports and timers. |

## 4. Architecture diagram

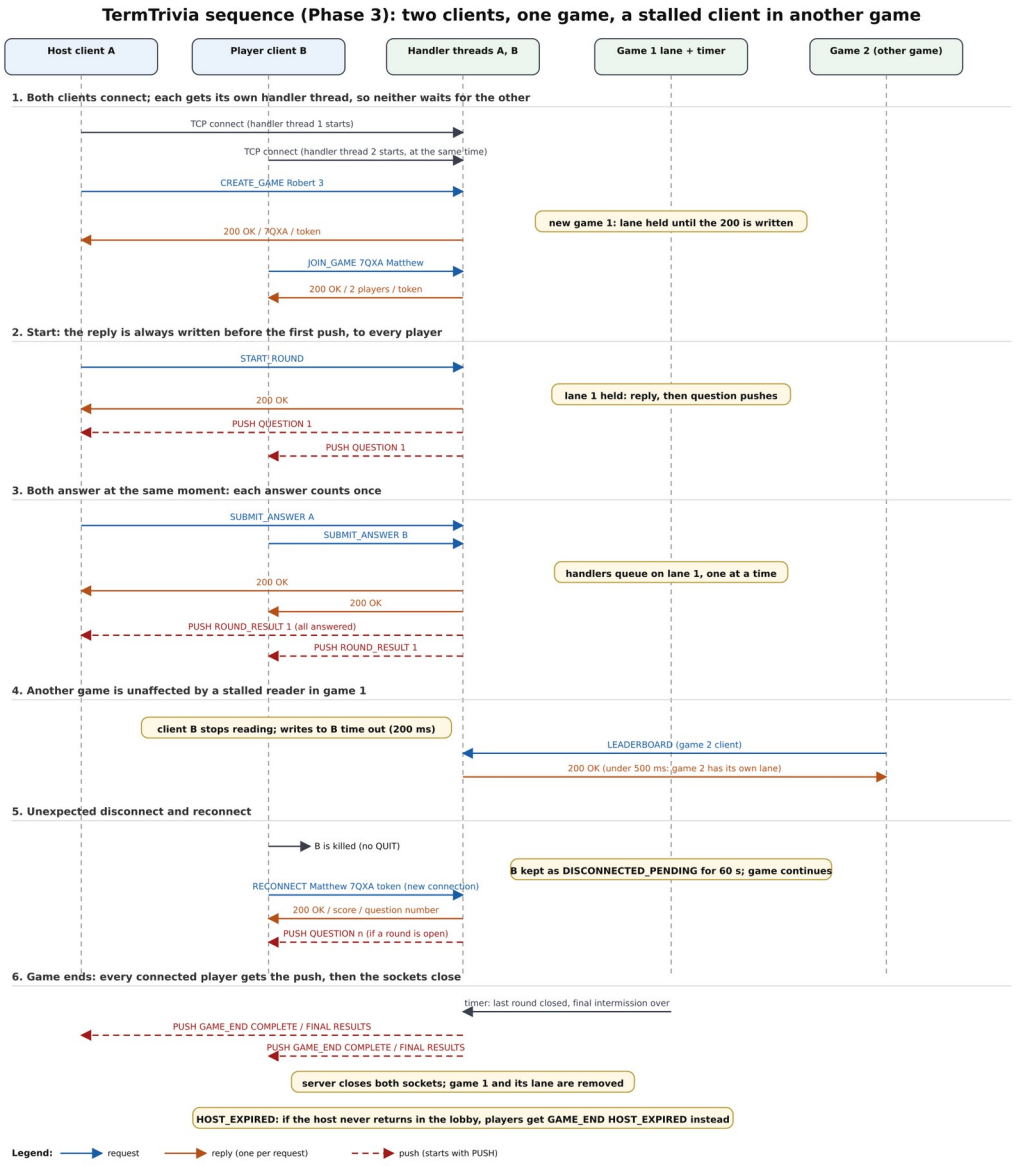
Source: `docs/images/architecture_diagram.svg` (PNG and PDF copies alongside).




Each client has a main thread (prompt), a reader thread (the only caller of recv), and a reply queue. The server has an accept loop, one handler thread per client, one timer thread per game, a lock ("lane") per game, a short routing lock, the game-state object, the logger, and the question bank.

## 5. Sequence diagram

Source: `docs/images/sequence_diagram.svg`.




The diagram follows two clients through connecting, creating and joining, starting, simultaneous answers, a stalled reader in another game, a disconnect and reconnect, and the end of the game.

## 6. Concurrency design

**Threads.** The accept loop starts one handler thread per connection; the handler owns that connection and is the only reader of its socket. Each game has one timer thread that ticks only that game every 20 ms. Writes can come from a handler (replies, and pushes caused by that command) or from the timer thread (question, result, and end pushes).

**Per-game lane.** The server first used one global lock for all games, which kept messages in order but let one slow client delay everyone. With three clients in one game not reading, a reply in another game took longer than 500 ms, and the test T40 failed on that design (5 of 6 runs). The final design gives every game a lane, a mutex that is held from the state change until all of that event's messages are written. That keeps **reply-before-push** and the order of events inside a game, while different games write at the same time.

**Lock order** is always: the game's lane (or the connection's own lane when it is not in a game), then the short routing mutex, then the game manager's own mutex. The routing mutex protects the game-code-to-lane map and the connection map. It may cover a state change and a registry update,

but never a socket call. Because every path uses the same order, threads cannot deadlock.

**Choosing the lane.** A command takes the lane of the game the connection is in; JOIN_GAME and RECONNECT use the code in the command. After locking, the server looks the lane up again, and retries if the game was removed or replaced, so a stale lookup never touches a newer game that reuses the code.

**Creating a game.** CREATE_GAME holds the new game's lane before the game is published, starts the game's timer thread, and only then writes 200 OK. If the timer thread cannot start, the game and lane are rolled back while the lane is still held, so a concurrent JOIN_GAME cannot enter a game that is about to disappear, and the connection is closed with no reply. This resource-exhaustion path is the one exception to the one-reply rule (T44).

**Delivery and lifetime.** Each effect is paired with a `shared_ptr` to its connection while the state change is protected, then written outside the routing mutex under the connection's own write lock with a 200 ms send timeout. A connection stays alive until the last pending write finishes, so no write can reach a closed or reused descriptor.

**Cleanup and shutdown.** Every worker sets a done flag as its last step. The accept loop joins finished workers after each wake-up; at shutdown it stops accepting, lets handlers and timers see the stop flag (handlers poll every 50 ms), and joins handlers first and timers second. If a handler thread cannot be created, that connection is closed and the error is logged while the server keeps accepting (T44).

**Host expiry.** The timer thread expires reconnect windows. An expired host in a lobby ends the game with GAME_END HOST_EXPIRED; during play the host is just removed (T42, T47).

## 7. Protocol specification

Full text: `docs/protocol.md`. The protocol is line-oriented ASCII over TCP. A client message is one line ending in a newline. A server message is one or more lines followed by a blank line; a reply is the answer to exactly one command and a push starts with `PUSH`.

| Command | Meaning and success reply |
|---|---|
| CREATE_GAME <name> <count> | 200 OK, join code, private token |
| JOIN_GAME <code> <name> | 200 OK, “N players in game CODE”, token |
| START_ROUND (host) | 200 OK, then PUSH QUESTION 1 to every active player |
| SUBMIT_ANSWER <A-D> | 200 OK (correctness is revealed in ROUND_RESULT) |
| LEADERBOARD | 200 OK, then name:score lines, highest first |

| | |
|---|---|
| RECONNECT <name> <code> <token> | 200 OK, “score question-number”, and the open question again |
| QUIT | 200 OK, then the connection closes |
| END_GAME (host) | 200 OK, then GAME_END HOST_ENDED to everyone |

**Pushes:** `PUSH QUESTION n text` with options A to D; `PUSH ROUND_RESULT n` with the correct answer and standings; `PUSH GAME_END reason` with final results. **End reasons:** COMPLETE, HOST_ENDED, HOST_QUIT, and HOST_EXPIRED (new in Phase 3). **Error codes:** 300 format, 401 not in a game, 402 host only, 404 unknown code, 409 duplicate name, 410 reconnect refused (one generic reply for any wrong detail), 420 no open round, 421 already started, 422 already answered, 423 bad question count. Phase 3 additions are recorded in `docs/design-changes.md` (decisions 38 to 43).

## 8. Data structures

| Structure | Where | Purpose |
|---|---|---|
| `Game` | game.h | code, state (Lobby, QuestionActive, Intermission), players in join order, shuffled question order, current question number, phase end time |
| `Player` | game.h | name, private token, score, state (Active, Waiting, DisconnectedPending), connection id, host flag, answered flag and choice, time of disconnect |
| `games_`, `game_code_of_conn_` | GameManager | games by code; which game a live connection is in (both under the manager mutex) |
| `Effect`, `Outcome` | game.h | game logic returns messages to send (connection, lines, close-after), log events, and created or ended game codes; it never touches sockets |
| `lanes_`, `live_connections_` | Server | game code to lane; connection id to shared connection (under the routing mutex) |
| `Worker` lists | Server | handler and timer threads with done flags, so finished threads are joined |
| `Connection`, `LineBuffer`, `MessageAssembler` | common/ | socket with a write lock and send timeout; bytes to lines (512-byte limit); lines to messages |
| `ReplyQueue` | client/ | hands replies from the reader thread to the prompt thread |

There is no persistent storage except the question bank file; games exist only in memory.

## 9. Error-handling strategy

- **Bad input:** every empty, unknown, malformed, or over-long line gets one error reply and the connection stays open. Input is data only and is never executed.
- **Wrong state:** specific codes (401, 402, 404, 409, 420 to 423) say what was wrong; no state changes.
- **Reconnect secrecy:** every wrong name, code, token, or late reconnect gets the same 410.
- **Slow or dead clients:** writes time out after 200 ms and mark the connection broken; the handler treats it as an unexpected disconnect. Resets and closes are handled the same way; SIGPIPE is ignored and sends use MSG_NOSIGNAL.
- **Disconnects:** the player is kept for the reconnect window and does not hold up a round; the server keeps running.
- **Host never returns:** a lobby ends with HOST_EXPIRED; a started game continues without a host.
- **Resource failure:** a handler thread that cannot start closes that client; a timer thread that cannot start rolls back the new game with no reply. Both are logged and the server keeps accepting.
- **Shutdown:** Ctrl+C stops accepting, ends every worker within about 50 ms, joins them, and logs one line.
- **Logging:** connections, requests with result codes, errors, game events, disconnects, and shutdown are logged; tokens never are.

## 10. Test results

Full detail, environment, and limitations: `docs/phase3-test-report.md`. Results were produced on one machine (Omarchy Linux, g++ 16.2.1, 8 CPUs) over localhost by the assistant that wrote the code; the team has not yet re-run them.

| Measure | Result |
|---|---|
| Unit tests (`make test`) | 116 tests run, 0 check failures |
| Real-socket tests | 75 tests run, 0 check failures |
| Scripted scenarios and deliverable audit | all checks passed |
| Real 20/5/10/60 second timers | all checks passed (`make test-default-timers`) |
| New Phase 3 socket tests (T10, T11, T22, T23, T35 to T47) | all pass; each repeated 10 times with 150 of 150 runs passing |
| 20 players, real timers (NFR-01) | reply 1 ms (limit 500 ms), question push 2 ms (limit 1 s), 20-second round timer 20 001 to 20 006 ms (limit plus or minus 1 s) |
| ThreadSanitizer, 75 socket tests | no report in Phase 3 code; one report in a Phase 2 startup test |

| | |
|---|---|
| | (not changed) |

**What the key tests showed.** Simultaneous answers each counted once with correct scores (T36); a same-name join race gave one 200 and one 409 (T37); ending game A left game B running (T38); three stalled readers in one game did not delay another game's replies beyond 500 ms (T40), while the earlier single-lock design failed that test in 5 of 6 runs; a lobby with an absent host ended with `HOST_EXPIRED` and a started game ran on to `COMPLETE` (T42, T47); a join in the final intermission appeared in the final results (T43); and failed thread launches, shutdown, and thread cleanup behaved as designed (T44). Testing also found one real bug, a reply that could be written after `GAME_END`; it was fixed and covered by a regression test.

**Limits of this evidence:** one machine, localhost only, at most 20 players, and timing bounds that assume a lightly loaded machine.

## 11. Known limitations

- One thread per client and per game: simple, and sound for the tested scale, but not for thousands of clients.
- A client that stops reading costs its own game up to 200 ms of waiting per write before it is dropped.
- No host transfer; no persistence; all games are lost if the server stops.
- No encryption and no user accounts; reconnect safety rests on a four-character private token. All testing used localhost and one machine, never a real network, never more than 20 players.
- A terminal and typing are required; a question still closes after 20 seconds.
- The client lists a join code but cannot browse games; there is no chat, team mode, or custom timer per question.

## 12. Scalability analysis

**Design.** The server uses one thread per connected client and one timer thread per active game. Per client the cost is a thread stack, a socket, and small buffers; per game it is a thread, a lane, and a few hundred bytes of state. Within a game all events are handled one at a time under that game's lane; different games run in parallel.

**Measured.** With 20 healthy localhost clients in one game and the real timers, the leaderboard reply took 1 ms, the START_ROUND reply 1 ms, the slowest client saw question 1 within 2 ms, and the 20-second round timer closed within 6 ms of 20 seconds. Three clients that stopped reading in one game did not delay another game's reply beyond 500 ms (T40). These meet NFR-01 for up to 20 players on one machine.

**Where it would stop scaling.** (1) Threads and memory grow with clients and games, so a few thousand connections would be limited by threads, stacks, and context switching. (2) A broadcast writes to each player in turn inside the game's lane; with many players or slow readers one game's round could be delayed by up to 200 ms per stalled reader. (3) Everything lives in one process on one machine, with no persistence or failover. (4) The game-state mutex is shared by all games, but it is held only for short in-memory operations, never for socket I/O.

**What would come next.** An event loop or a worker pool, a per-connection outbound queue with a writer so a slow reader never delays a broadcast, and sharding games across processes. We did not measure 25 or more players, many simultaneous games, or a real network, so no claim is made about them.

## 13. Team contribution summary

### Who did what

| Phase | Robert | Matthew |
|---|---|---|
| 1 | Curiosity report, project proposal, architecture diagram, protocol specification, data design | Functional and non-functional requirements, testing plan |
| 2 | Set up the AGENTS.md workflow and used AI coding agents to develop, write, and test the code | Tested the server with a single client (did most of the manual testing) |
| 3 | Used the same workflow to develop and test the multi-client code | Tested the multiple-client behavior manually |
| Code review and demo preparation | Reviews and explains the server code | Reviews and explains the client code |

The Phase 3 technical report, diagrams, and draft documentation were generated by the AI agents from the team's specifications. The team wrote this section itself.

### Robert: individual reflection

**My contributions.** In Phase 1 I wrote the curiosity report, the project proposal, the architecture, the protocol specification, and the data design. In Phases 2 and 3 I set up the agent workflow and directed the agents that wrote and tested the code. I am responsible for understanding and explaining the server code in the demonstration.

**Most important networking concept.** Handling a client that disconnects. I learned how a private token ties a player to their seat, so when the player reconnects they get back their own score and nobody else can take their place.

**Technical challenge.** Deciding what should happen when a player drops. Should they be removed completely, or can they come back, and how do we stop someone else from taking the seat? The two problems were linked. We settled on a private token plus a reconnect window: a dropped player keeps their place for 60 seconds by default, and a connection without the token cannot claim it.

**Skill to improve.** My understanding of socket programming. I had not done it before this project. I used the agents to build the code, then reviewed it myself and used AI to explain what it does. I learned a lot, and there is a lot more to learn.

**How my work contributed to user value.** I defined the core ideas the agents built from: scoring

is based on correct answers and not speed, so the game is fair, and players who disconnect can rejoin without losing their place. I gave the agents these ideas and made sure the code followed them.

### Matthew: individual reflection

**Pending student input.** Matthew must write his own contributions, the most important networking concept he learned, a technical challenge he overcame, a skill he wants to improve, and how his work contributed to user value.

## 14. Ripple Report reflection

**Pending student input.** The team must write its own Ripple Report reflection. No reflection has been written or implied in this draft.